

POWER SUPPLIES



Power supply

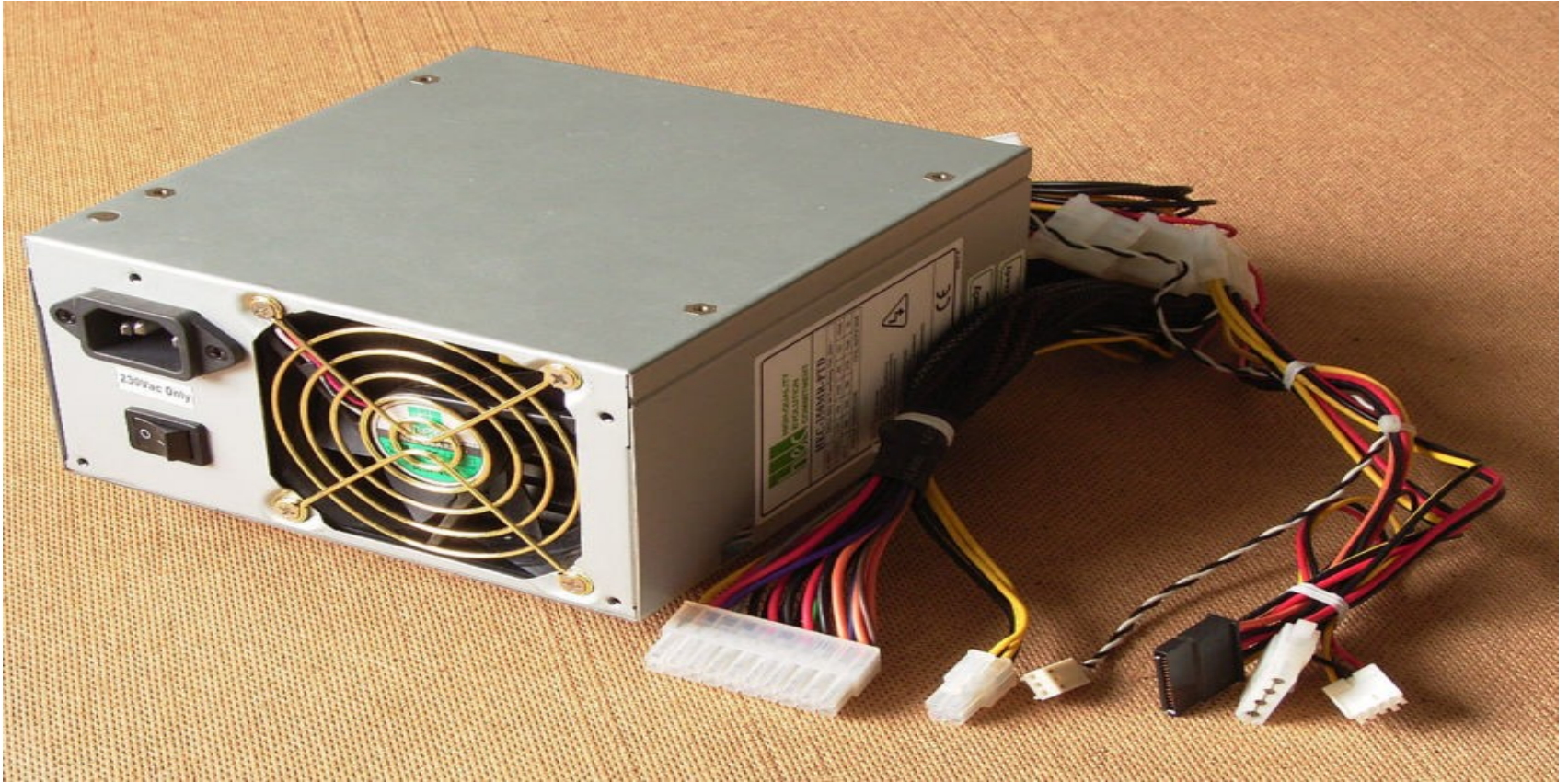
- ☑ Converts AC power from the wall socket into DC
- ☑ Must provide enough power for the installed components and future additions

Power Supplier

The basic function of power supply is:

- ☑ Supplying power to the system
- ☑ **Convert** the type of electrical power available (Alternate Current-AC) to the type computer circuitry can use (Direct Current).
- ☑ Ensures that the system does not run unless power supply is **sufficient** to operate the system **properly** (the power supply actually prevents the computer from running (starting up) or operating until all correct power levels are present, Each power supply completes **internal checks** and tests before allowing the system to **start**. If tests are successful, the **power supply sends a special signal called Power Good to the motherboard**. If this signal not present, computer does not run.)

Power Supplier...



Power Supplier...

☑ Fact:

- ☑ Almost everything in your home runs off standard AC power.
But they actually use DC internally.

E.g. printers, external modem, scanner, etc.

Attention!:

- ☑ plugging the power connectors in the wrong direction could damage the computer, Pay attention when you plug.

Connectors to Peripheral Hardware

- ☑ **Three standard types of connectors are used to connect power to peripheral devices:**
 - ☑ **Molex connectors**
 - ☑ **Mini connectors**
 - ☑ **SATA connectors**

Connectors to Peripheral Hardware...

1. Molex Connector

- ☑ This was the most commonly used connector in old PCs.
- ☑ Older Hard disk drives, internal tape drives, CD-ROM drives, DVD drives, and 5^{1/4} floppy drives used this power connector.
- ☑ It has **four pins**.
- ☑ It also exists in some of today's PSUs for backward compatibility.
- ☑ It has a 5V and 12V power supply, and ground wires for each.

Connectors to Peripheral Hardware...

2. Mini Connector

- ☑ It is used primarily for 3.5 floppy disk drive.
- ☑ It has **four pins** and **four wires**.
- ☑ Connecting the connector in opposite direction could damage the system.
- ☑ The good thing is it does not fit in the wrong direction

Connectors to Peripheral Hardware...

3. SATA Connector

- ☑ This is the most commonly used connector.
- ☑ Hard disk drives, internal tape drives, CD-ROM drives, DVD drives, and older 5^{1/4} floppy drives use this power connector.
- ☑ It has **four pins**.
- ☑ For modern computers, SATA is the common protocol, and most new drives use a SATA power connector, so the Molex connector is becoming obsolete.

Cont'd



Power Failure

- ☑ Power supplies are affected by **quality** of local power source.
- ☑ **Common power delivery problems are:**
 - ☑ Spikes
 - ☑ Surges
 - ☑ Sags
 - ☑ Brown outs & black outs

Power Failure...

1.Surge

- ☑ A brief increase in the voltage source .

2.Spikes

- ☑ Are very short over voltage conditions.

3.Sags

- ☑ A brief decrease of voltage at power source.

4.Brown Outs

- ☑ If a sag lasts longer than 1 second, it is called brown out.

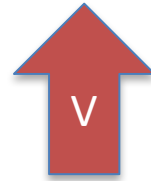
5.Black Outs

- ☑ This is a complete power failure.
- ☑ This could be caused by equipment failure, or accidental cutting of power cable.

Power Failure...

High

- **Spike:** Temporary High
- **Surge:** Prolonged High



Low

- **Sag:** Temporary Low
- **Brown Out:** Prolonged Low



Loss

- **Fault** : Temporary Loss
- **Black out:** Prolonged loss

The general power problems can be categorized in three types:

1) Power Quality problem

Possible causes:

- ☑ EMI (Electro-Magnetic Interference) .
- ☑ Variable rate [frequency] from the power line.,

2) Too much power

Possible causes:

- ☑ Power spike [for few milliseconds].
- ☑ Power surge [for several seconds].

3) No-enough power

Possible causes:

- ☑ Power sag [for few milli second]
- ☑ Brownout [if sag lasts for longer than a second]
- ☑ Blackout [a complete of power failure]

Protection against power problems

You can use the following devices

- ☑ Line conditioner
- ☑ Stabilizer
- ☑ Surge suppressor
- ☑ UPS

☑ N.B:

- ☑ ups means uninterruptible *Power Supply*.
- ☑ The UPS is also known as battery backup

Power Control

There are different devices that enable us to control power.

a) UPS (Uninterruptible Power Supply):

- ☑ A UPS helps in **black out** situations when the power is gone totally and **brown outs**, when voltage is low though there is power.
- ☑ UPS has batteries that provides back up power.
- ☑ When the power is gone, the battery provides voltage to the PC so that you can save your work.
- ☑ But it **does not provide unlimited power**.

Power Control...

There are two types of UPS:

1. Standby UPS:

- ☑ SPS has a battery that begins generating power as soon as the unit detects a **sag(fall)** in the power supply.
- ☑ It takes only a split of seconds to come online.
- ☑ The disadvantage is that till the UPS become online, your **data might get lost**.

Power Control...

types of UPS:

2. Online UPS:

- ☑ It provides electricity to the PC **all the time**.
- ☑ It uses electricity from the AC outlet to simply recharge its batteries.
- ☑ When the power goes, the **data is not affected** because the UPS is supplying power.

Power Control...



Fig UPS (front and back view) and fuse

Power Control...

b) Surge Suppressor:

- ☑ help to absorb power surges(flows), so that your computer does not feel their effects.



Fig: Surge suppressor

Power Control...

c) Voltage Stabilizer

- ☑ It is a transformer that delivers relatively constant output when output voltage changes over time.
- ☑ The output voltage is regulated using transistor.

Power Control...

D) CMOS Battery

- ☑ Computer has a Chip that combines real time clock and non-volatile memory.
- ☑ This chip is the CMOS chip (CMOS RAM).
- ☑ They are designed to consume low power.

Battery

- ☑ The CMOS chip stores basic system configuration like amount of memory installed, type of floppy and hard disk drives, type of keyboard and mouse, etc.
- ☑ This is powered by battery when system is off.
- ☑ This battery keeps the time chip running and the CMOS powered.
- ☑ Most systems use **lithium type battery** because they have very long life.

Battery

- ☑ Most of the higher quality systems sold today have a new type of chip that has the battery embedded in it.
- ☑ These chips are notable for long life. Under normal conditions, the battery lasts for **10** years.
- ☑ Most of the time, these chips-battery combinations are installed in a socket on the motherboard.
- ☑ Some systems do not use battery at all.

Troubleshoot Power Supply

PC problems that often relate to PSU are:

- ☑ Any power-on or system startup failures or lockups
- ☑ Spontaneous rebooting or intermittent(irregular) lockups during normal operation
- ☑ Intermittent parity (uniform) check or other memory-type errors
- ☑ Hard disk and fan simultaneously failing to spin

Troubleshoot Power Supply

PC problems that often relate to PSU are:

- ☑ Overheating due to fan failure
- ☑ Small brownouts that cause the system to reset
- ☑ Electric shocks/wave felt on the system case or connectors
- ☑ Slight/small static discharges that disrupt system operation

Cont...

Problem

- Monitor's power indicator lights but no power lights on the system unit.

Solution

- Check the system unit's power connection.
- Check your power supply DC volt outlet.
- Use a replacement method for power supply.

Power supply symptoms

- Fan noise sounds rough or louder than usual.
- Fan noise is absent altogether.
- The power supply chassis is unusually hot to touch.

Cont...

General troubleshooting methods

- Check the power cable.
- Check the power supply [you can use a Multi- meter to test electronic components].
- Check the power outlet voltage.
- Replace the power supply unit.

Problem

- A PC accidentally reboots or shutdown

Solution

- Check your power line.
- Make sure that your power supply is rated [watt" to handle all the peripherals that it powers [300 watt or above is better]
- Check the power supply fan movement.
- Use a replacement method.

Thank You

End of the Chapter